Physical Chemistry Cume: SPECTROSCOPY Feb 3 2017

1. Describe the physical origins of linewidths in absorption and emission spectra. Explain how their relative contributions change between the gas and condensed phase. What can you do to reduce spectral linewidths?

2. The force constant of a HCl molecule is 516.6 Nm^{-1} and the bond length is 127.5 pm
   Atomic masses: ^{35}Cl = 34.9689 a.m.u., H = 1.0078 a.m.u., D = ^{2}H = 2.014 a.m.u.
   (a) What **wavelength** of light is needed to excite (i) $H^{35}Cl$ and (ii) $D^{35}Cl$ from the first excited rotational state to the second excited rotational state?
   (b) What **wavelength** of light is needed to excite (i) $H^{35}Cl$ and (ii) $D^{35}Cl$ from the ground vibrational state to the first excited state?
   (c) If an infrared spectrum of gaseous $H^{35}Cl$ is measured, what is the **wavenumber** of the line in the R branch that would correspond to the transition originating from J = 2. Ignore any anharmonicity and centrifugal distortion effects.
   HINT: make sure you convert your answers from part (a) and/or (b) into wavenumbers first.

Useful Information:

c = 2.9979 × 10^{8} ms^{-1}

h =6.62608 × 10^{-34}Js or $\hbar$ =1.05457 × 10^{-34}Js

1 Atomic mass unit (a.m.u) = 1.66054× 10^{-27} kg

Some useful equations:

$$E_J = hBJ(J+1)$$

$$B = \frac{\hbar}{4\pi I}$$

$$\tilde{B} = \frac{B}{c}$$

for a diatomic molecule: $I = \mu r^2$

$$E_v = \left(v + \frac{1}{2}\right)\hbar\omega$$

$$\omega = \sqrt{\frac{k_f}{\mu}}$$

$$\tilde{\upsilon}_R(J) = \tilde{\upsilon}_0 + 2\tilde{B}(J+1)$$